Day 1 – Dataset Exploration

Data Handling and Machine Learning | Ghanshyam Bhattarai | 18 August 2026

Row meaning: Each row looks like one student. It shows the student's study method, Python confidence, study hours, completed tasks, quiz score, last activity and group. However, the dataset does not clearly say whether this information is from one week or the whole course.

Variable types: `student_key` is the student ID. `study_mode` and `group_code` are categorical variables. `python_confidence` is ordinal because low, medium and high have a clear order. `weekly_study_hours` is continuous numerical data, while `tasks_completed` is a discrete count. `quiz_score_pct` is numerical, and `last_activity` is a date.

Data-quality problems: Some values are missing: the quiz score for SYN012, group code for SYN015 and study hours for SYN020. Online is written in different ways, and the dates use three formats. There are also impossible or questionable values such as `-4` study hours and a `104%` quiz score. The word `twenty` is used instead of a number, and SYN023 appears twice.

Meaningful summaries: I could count the number of students in each study mode, confidence level and group. I could also calculate the average or median study hours and quiz scores, count completed tasks and find the earliest and latest activity dates. Calculating the average of student IDs or group codes would not make sense.

Reasonable questions: This data could show whether students who study more or complete more tasks usually get better quiz scores. It could also help compare results between Campus, Online and Hybrid students. However, it cannot prove that one study mode causes better results because we do not know how the students were chosen or what other factors affected them.

Information needed: I would want to know what period the study hours cover, how the hours and confidence levels were collected, the total number of tasks and whether quiz scores can go above 100. I would also ask what the group codes and missing values mean and whether the dataset is real or made for practice.

First preparation decisions: First, I would check whether SYN023 is an accidental duplicate before removing it. Second, I would check whether `twenty` means 20, why one student has `-4` hours and what the missing study-hours value means. I would confirm these things before changing the data.